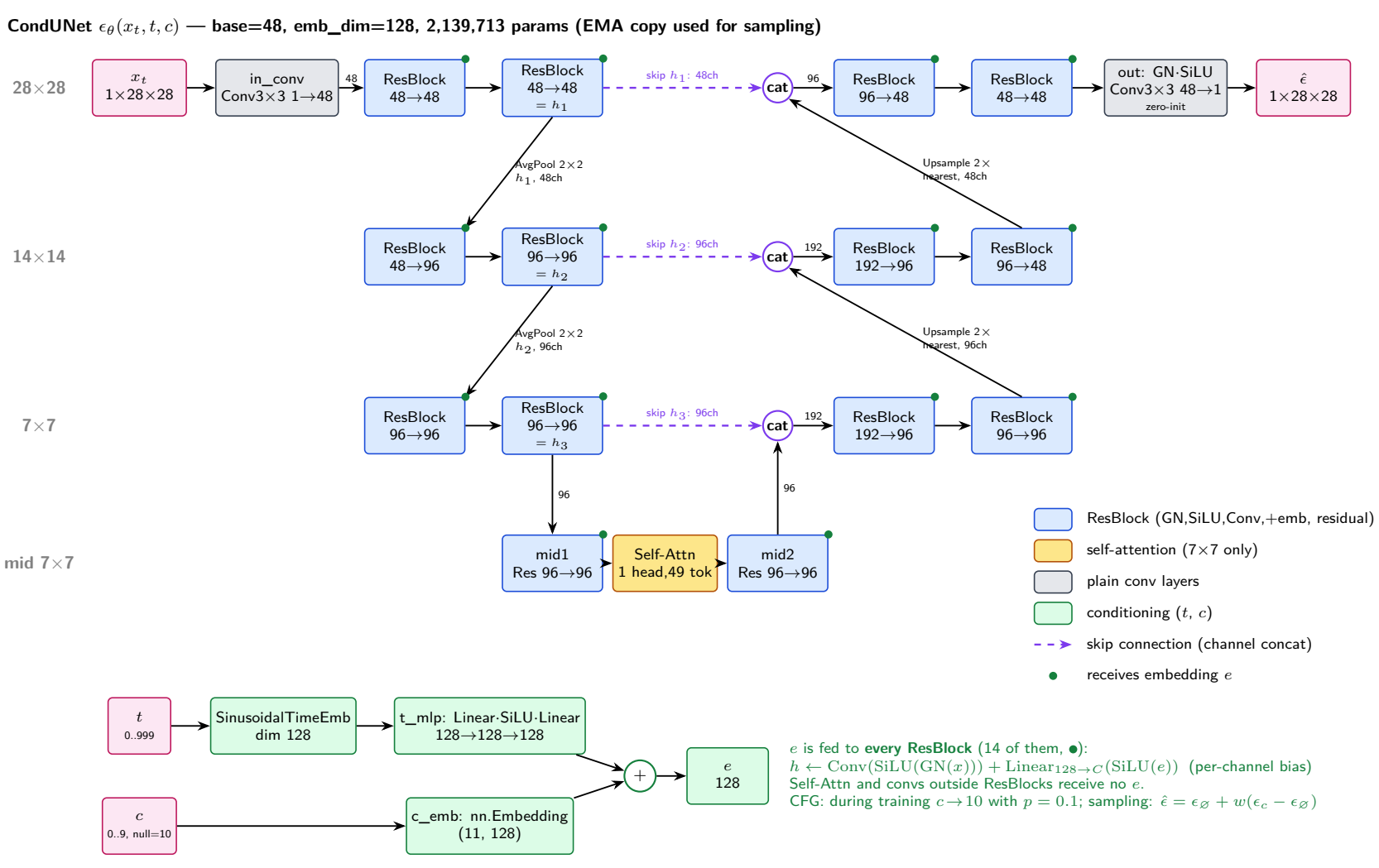
CondUNet $\epsilon_\theta(x_t, t, c)$ — base=48, emb_dim=128, 2,139,713 params (EMA copy used for sampling)
28×28
14×14
7×7
mid 7×7
x_t 1×28×28
in_conv Conv3×3 1→48
48
ResBlock 48→48
ResBlock 48→48 = h_1
skip h_1: 48ch
cat
96
ResBlock 96→48
ResBlock 48→48
out: GN·SiLU Conv3×3 48→1 zero-init
$\hat{\epsilon}$ 1×28×28
AvgPool 2×2 h_1, 48ch
Upsample 2× nearest, 48ch
ResBlock 48→96
ResBlock 96→96 = h_2
skip h_2: 96ch
192
ResBlock 192→96
ResBlock 96→48
AvgPool 2×2 h_2, 96ch
Upsample 2× nearest, 96ch
ResBlock 96→96
ResBlock 96→96 = h_3
skip h_3: 96ch
ResBlock 192→96
ResBlock 96→96
96
96
mid1 Res 96→96
Self-Attn 1 head,49 tok
mid2 Res 96→96
ResBlock (GN,SiLU,Conv,+emb, residual)
self-attention (7×7 only)
plain conv layers
conditioning (t, c)
skip connection (channel concat)
receives embedding e
t 0..999
SinusoidalTimeEmb dim 128
t_mlp: Linear·SiLU·Linear 128→128→128
c 0..9, null=10
c_emb: nn.Embedding (11, 128)
e 128
e is fed to every ResBlock (14 of them, ●):
$h \leftarrow \mathrm{Conv}(\mathrm{SiLU}(\mathrm{GN}(x))) + \mathrm{Linear}_{128\to C}(\mathrm{SiLU}(e))$ (per-channel bias)
Self-Attn and convs outside ResBlocks receive no e.
CFG: during training $c \to 10$ with $p = 0.1$; sampling: $\hat{\epsilon} = \epsilon_\varnothing + w(\epsilon_c - \epsilon_\varnothing)$